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PlayList : DevOps with AWS full course

channel : India Free Internship

notes Link : https://github.com/indiafreeinternship/NOTES/tree/main/spring%20boot%20notes

live notes and live coding

https://www.youtube.com/@IndiaFreeInternship

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session 1 ppt

DevOps Introduction

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session 2

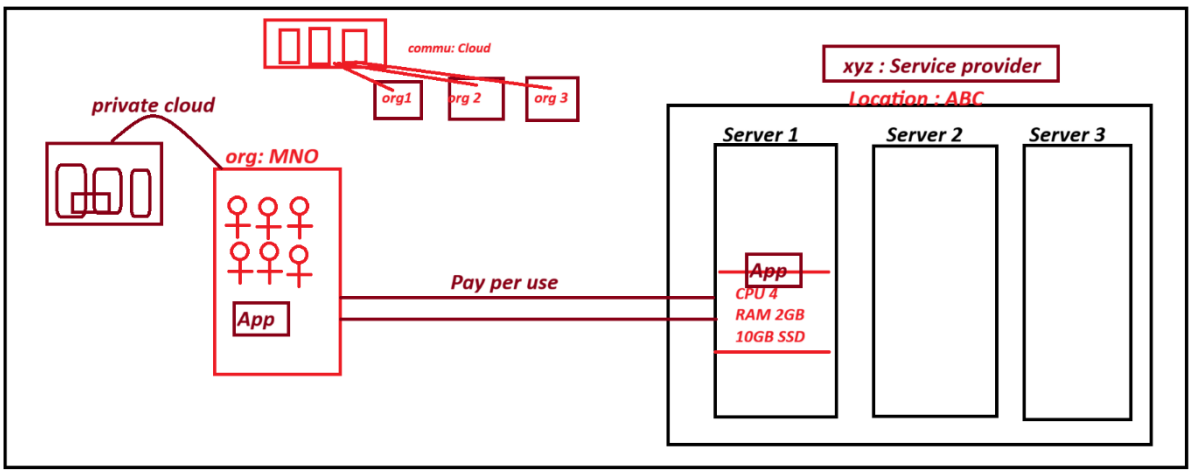
AWS (Amozon Web Services)

Introduction

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1. Cloud Computing :

Cloud comutin is the on-demand abailability of computer system resources, especially data storage and computing power, without direct active management by the user. Large often thave functions distributed over multiple locations, each of which is a data center



diagram

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2. Cloud providers

The big three -

1. Google Cloud
2. Microsoft Azure
3. AWS (Amazon Web Services) -

are considered the established leaders. However, there are a host of other smaller or niche players that offer cloud services as well, including IBM, Alibaba, Oracle, Red Hat, Digital ocean, and Rackspace.

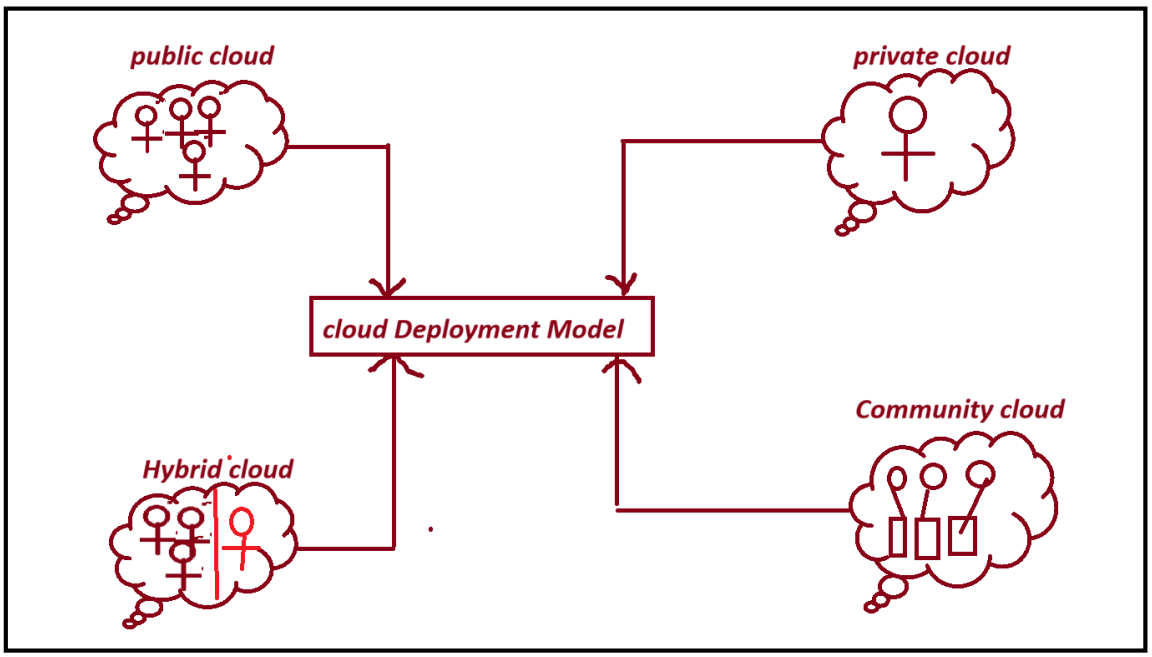
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3. Cloud Computing Deployement Models:

The cloud deployment model indentifies the specific type of cloud environment based on ownership scale, and access, as well as the cloud's nature and purpose. The location of the servers you're untilizing and who controls them are defined by a cloud deployment model. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself. Relationships between the infrastructure and your users are also defined by cloud deployment types.

Different types of cloud computing deployment models are :

1. Public cloud
2. private Cloud
3. Hybrid cloud
4. Commnunity cloud



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1. Public cloud:

The public cloud makes it possible for anybody to access systems and services.

The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups. It is a type of cloud hosting that allows customers and user

to easily access systems and services. In this arrangment, storage backup and retrieval services are given for free, as a subscription, or on a per-user basis. Example : Google App Engine etc.

Advantages of public cloud Model:

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(i) Minimal investment: Because it is a pay-per-use service, there is no sustantial upfront fee, making it excellent fro enterprises that require immediate access to resources.

(ii) No Setup cost: The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.

(iii) Infrastructure managment is not required: Using the public cloud does not necessitate infrastructure managment.

(iv) No maintenance: The maintenance work is done by the service provider (Not users).

(v) Dynamic Scalability : To fulfil your company's needs, on-demands resources are accessible.

DisAdvantages of public cloud Model:

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Less Secure : public cloud is less secure as resources are public so there is no guarantee of high level security.

Low customization: It is accessed by many public so it can't be customized according to personal requirement.

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2. private Cloud : It's a one-on-one environment for a single user (customer). There is no need to share your hardware with anyone else. It is also called the "Internal cloud" & it refers to the ability to access systems and services within a given border or organization. The cloud platform is implemented in a cloud based secure enirоment that is protected by powerful firewalls and under the supervision of an organization's IT department.

Advantages of private cloud Model:

(i) Better Control: You are the sole owner of the property. You gain complete command over service integeration, IT operation, policies, and user behaviours.

(ii) Data Security and Privacy: It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.

(iii) Support Legacy systems: This approcate is designed to work with legacy stystems taht are unable to access the public cloud.

(iv) Customization : Unlike a public clood deployment, a private cloud allows a compnay to tailor its solution to meet its specific needs.

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DisAdvantages of private cloud Model:

Less Scalable: private cloud are scaled within a certain range as there is less number of clients.

Costly: private clouds are more costly as they provide personalized facilities.

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3. Hybrid cloud: By bridgin the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organization can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.

Advantages of Hybrid cloud Model:

(i) Flexibility and control: Bussiness with more flexibilit can design personilzed solutions that meet their particular needs.

(ii) cost: Because public clouds provide scalability, you'll only be responsible for paying for extra capacity if you required it.

(iii) Security: Bacause data is proporly separated, the chances of data theft by attackers are considerably reduced.

DisAdvantages of Hybrid cloud Model:

Difficult to manage: Hybrid clouds are dificult to manges as it is a combination of both public and private cloud. So it is complex.

Slow data transmission :Data Transmission in the hybrid cloud takes place through the public cloud so latency occurs.

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4. Commnunity cloud: It allows systems and services to be accessible by a group of organizations. It is distributed system that is created by integrating the services of different cloud to address the specific needs of a community, industry, or business. The infrastructure of the community cloud be shared between the organization which ahs shared concerns tor taks. It is generally manged by a third party or by the combination of one or more organization in the community.

Advantages of Community cloud model:

(i) cost Effective: It is cost effective becuase the cloud is shared by multiple organization or communities.

(ii) Security: community cloud provide high security.

(iii) Collaboration and data sharing: It is suitable for both collaboration and data sharing.

(iv) Shared resources: It allows you to share resources, instrastructure, etc with multiple organization.

Advantages of Community cloud model:

Limited Scalability: Community cloud model is relatively less scalable as many organization share the same resources according to their collaborative interests.

Rigid is Customization: As the data and resources are shared among different organizations according to their mutual interests if an organization watns some changes according to their needd they cannot do so becuase it will have an impact on other oraganizations.

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session 3

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4. Cloud Computing Services.

There are the following three tyes of cloud services models:

a. Infrastructure as a Service (IaaS)

b. Platform as a Service (PaaS)

b. Software as a Service (SaaS)

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a. Infrastructure as a Service (IaaS) :

It is also known as Hardware as a Services (HaaS).

It is a computing infrasture manged over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.

Characteristics of IaaS:

There are the following characteristics of IaaS.

1. Resource are avaiable as a service

2. Services are highly scalable

3. dynamic and flexible

4. GUI and API-based access

5. Automated administrative tasks

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b. Platform as a Service (PaaS) :

PaaS cloud computing platform is created for the programmer to develop, test, run, and manage the applications.

Characteristics of PaaS

There are the following characteristics of PaaS.

1. Accessible to various user via the same devlopement application

2. Integerated with web services and databases.

3. Builds on virturalization technology, so resources can easily be scaled up or down as per the organization's need.

4. Support multiple languages and framewoks.

5. provide an ability to 'Auto-Scale'.

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c. Software as a Services (SaaS):

It is also known "on-demand software'. It is a software in which the applications are hosted by a cloud service provider. user can access these application with the help of internet connection and web browser.

Characteristics of PaaS

There are the following characteristics of SaaS.

1. Managed from a central location

2. Hosted on a remote server

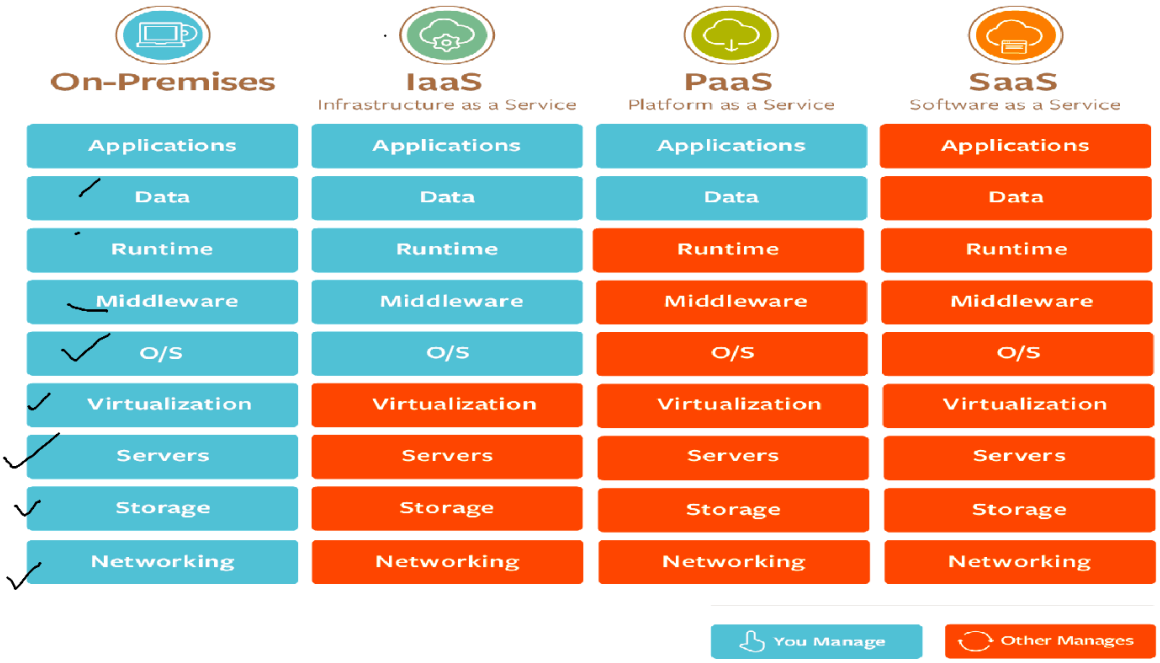
3. Accissible over the internet

4. user are not responsible for hardware and software updates. Updates are applied automatically.

5. The services are purchased on the pay-as-per use basis.

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Diagram



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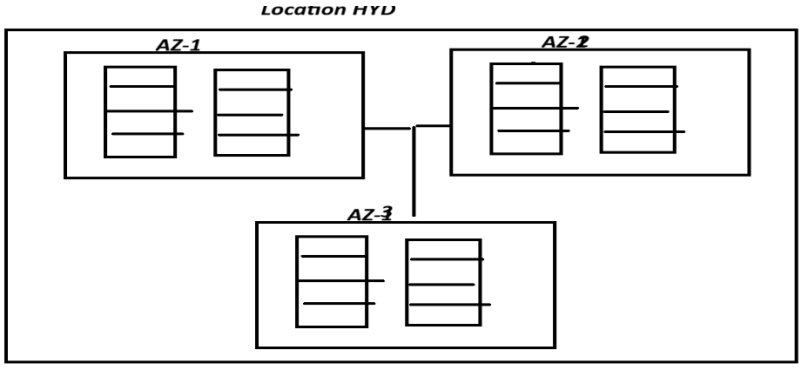
5. AWS:

AWS stands for Amazon Web Services which used distributed IT infrastructure to provide different IT resources on demand. It provide different services such as IaaS, PaaS, SaaS. Amazon launched AWS, a cloud computing platform to allow the different organizations to take advantages of reliable IT infrastructure.

Regions and Availability Zones:

An AWS regiion is a cluster of data centers in a specific geographic area, sucha s the Northeastern united States or Western Europe. It is a best practice to choose a region that is geographically close to users; this reduces latency becuase data reaches the user more quickly.

Each AWS Region includes mutiple AZs. However, each AZ is restricted to a specific AWS regions. You can use multiple AZs within one Region, but you can't use the same AZ across multple Regions.



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6. NIST's 5 Essential Cloud Computing Characteristics:

The National Institute of Standards Technology (NIST) list five essential characteristics of cloud computing: on-demand self-services, broad network access, resource pooling, rapid elasticity, and measured service.

a. On-Demand Self Services

b. Broad Network Access

c. Resource Pooling

d. Rapid Elasticity :

e. Measured Service:

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session 4

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Linux

Shell commands

Bash

GitHub/Git : Developer code (.html, .css, .java, ...etc)

Maven/Gradle: build tool (code -> packed)

Application/web server : Tomcat

(DevOps) Server : physical machine

(Developer/Dev) web server : Softaware that runs application

sonatype Nexus: Artficatory Repository (build files/ packages for distribution)

sonarqube : code quality checking.

[set of rules]

-code duplicated?

-code has no standard

-so many lines of code in single function/method?

-un-used variables?

Report : code smells, security issued..etc

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compiler : will check code syntax, convert into system/runtime understandable format.

ex Abc.java -> compile -> Abc.class (java Runtime)

UnitTest Code : Written by Developer but executed before package.

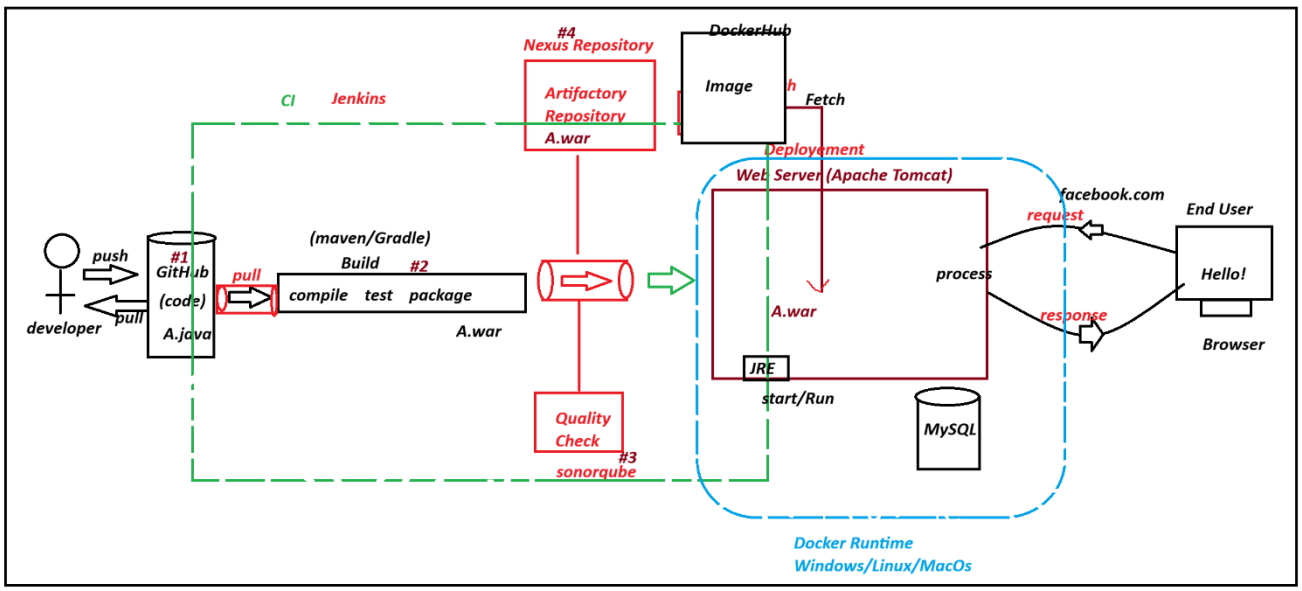
Repository : Memory/HardDisk that holds data.

code Repository : GitHub

Artifactory Repo: Nexus

Jenkins : CI/CD tools, it automates project delivery from source code to productions server an Autotmated pipe line is created, to deliver application to end customers.

Diagram



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Execution Flow

1. Developer writes source code and push into GitHub (code Repository/ Version control software)

2. By using a build tool (compile-> test-> package), code is converted into packaged formate.

ex: A.java -> compile->A.class->package->A.war(Web Archive)

Here maven/Gradle is used

3. For code quality cheching, sonarqube is used and validated, finally it generatd a report for a project showing all issued.

4. Build files also called as Artfiacts, such artifact must be stored in Artifactory Repository ex : SonaType nexus/JFrog(DockerHub, Aws ECR).

5. Finally web server (Tomcat, jetty, Undertow…etc) which runs applications from artifactory Repo. we need to copy build file (ex WAR file) and deploy (Deploy -place app in server and start server)

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Platform means OS (Runtime)

Software are platform depented.

Ex: java software, vlc mediaplayer it comes with different extensions.

MacOs -- .dmg

Linux -- .rpm

Windows --- .exe

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Docker

=>Docker is software used to make our application OS independent.

=> Docker Converts all our code and dependencies into single software called as Images.

These images can be stored into one Common place is called as DockerHub(similar to nexus)

Docker even contains runtime which need to install on top of OS. Then run images.

=> To run our application we need below componetns/softwares.

MyApp.war (1-Artifact)

[3-Dependencies]

JRE (java runtime)

Apache Tomcat webserver

MySQL Database

=> we want to run our application, on any OS by usnig Images concept.

Image S/w= OS independent (Artifact + dependencies) => MyApp

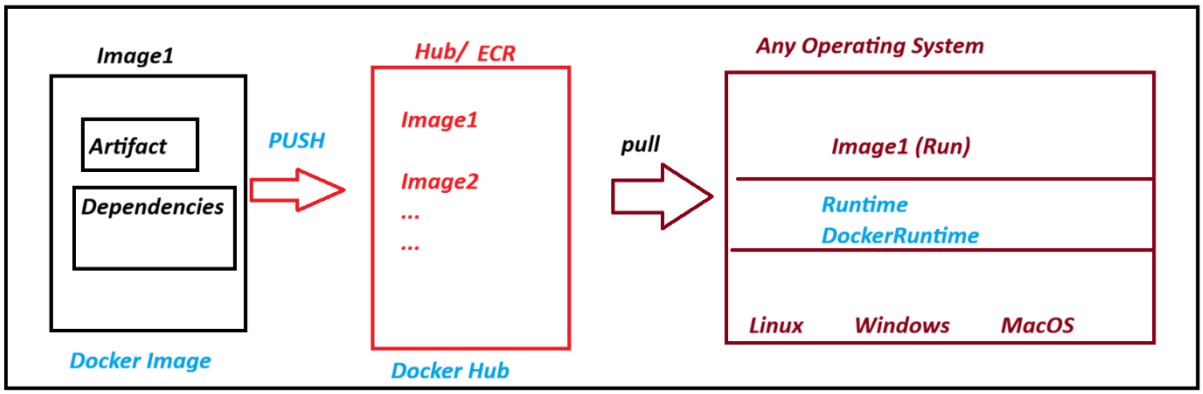
1. Create image

2. Ship Image (send from one machine to another machine) (push/pull) using Hub/ECR

3. Run image on any OS using Image Runtime.

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Diagram



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session 5